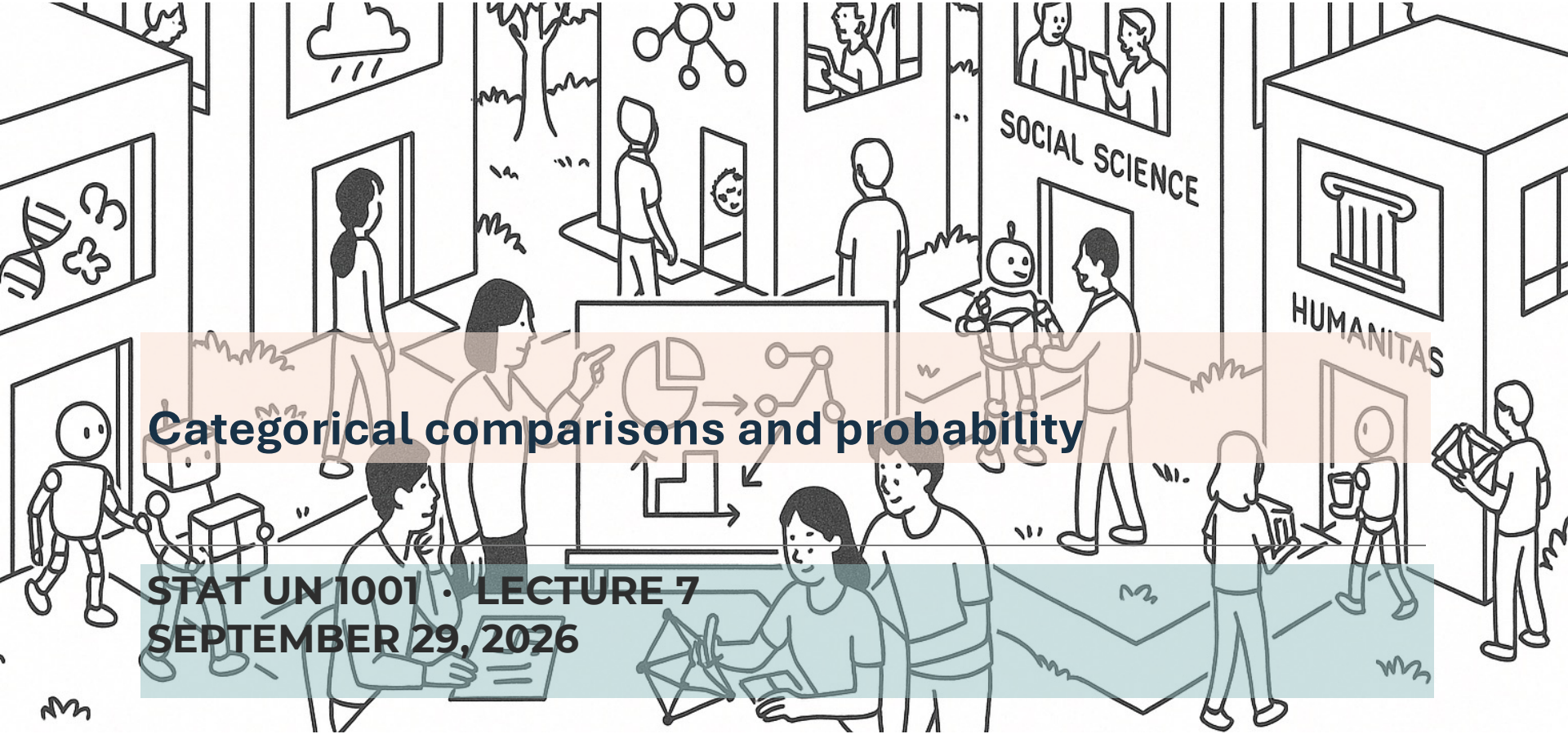


Categorical comparisons and probability

STAT UN 1001 · LECTURE 7
SEPTEMBER 29, 2026

From Flipped Activities Check-In

HARDEST ISSUE

Knowing which rule applies
26 students

NEXT

Understanding independence
13 students

PRACTICE QUIZ LOW POINTS

Independence from Venn: 45/64 correct
Two-city rain independence: 47/64 correct
Disjoint addition: 51/64 correct

Today’s repeated question: what does the event mean?

Counts and proportions

Report group	Event X	Other events	Total
Vaccine A	50	950	1,000
Older vaccines	90	8,910	9,000
Total	140	9,860	10,000

Which group has more reports?
Which group has the larger Event X proportion?

The comparison group

Report group	Event X	Total	Event X proportion
Vaccine A	250	5,000	5.0%
New Vaccine A2	225	4,500	5.0%
Older vaccines	5	500	1.0%

If “everyone else” mostly means New Vaccine A2, what question does the comparison answer?

Probability: Chance process and sample space

CHANCE PROCESS

Randomly sample one household

OUTCOME

one selected household's combination of traits

EVENTS

C: has a cat
A: has a car
B: has a child under age 2
...

SAMPLE SPACE

all possible combinations that could be selected

What outcome did we observe if the household has a cat, no car, and no child under 2?

Events as sets of outcomes

C AND A

cat and car

C OR A

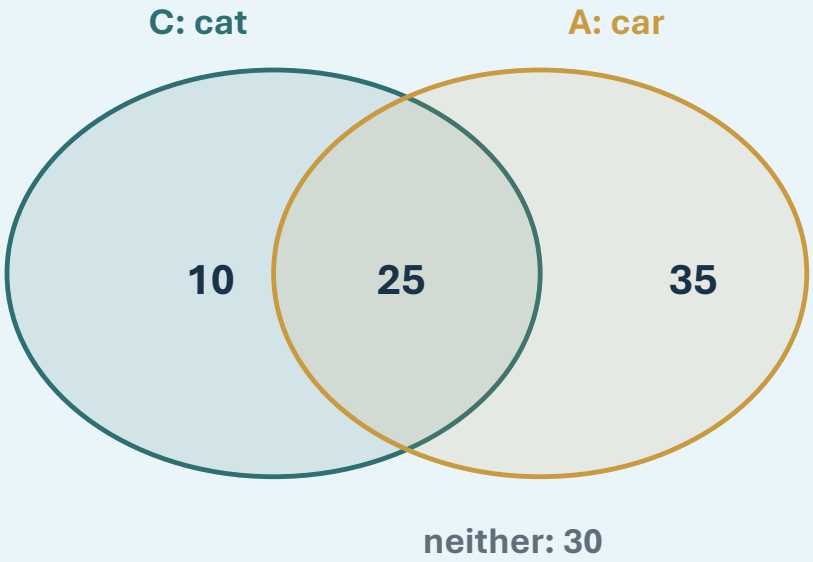
cat, car, or both

NOT C

no cat

In probability, “or” includes the overlap unless the problem says otherwise.

Venn diagram regions



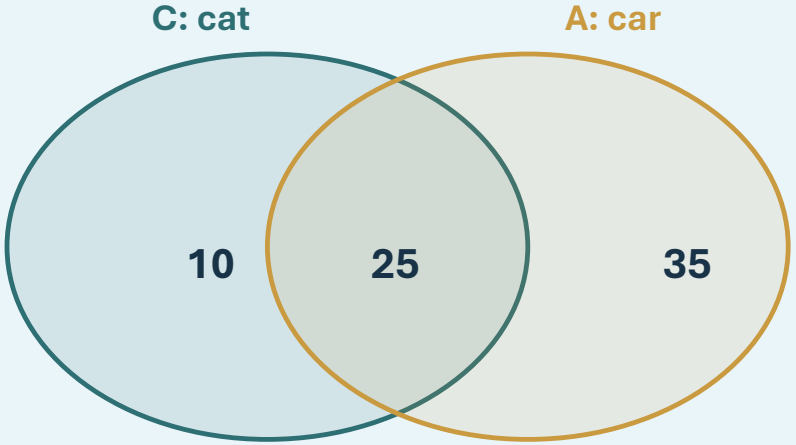
SURVEY 100 HOUSEHOLDS

35 have a cat
60 have a car
25 have both

READ FROM THE DIAGRAM

cat only = 10
both = 25
car only = 35
neither = 30

Overlap and double-counting



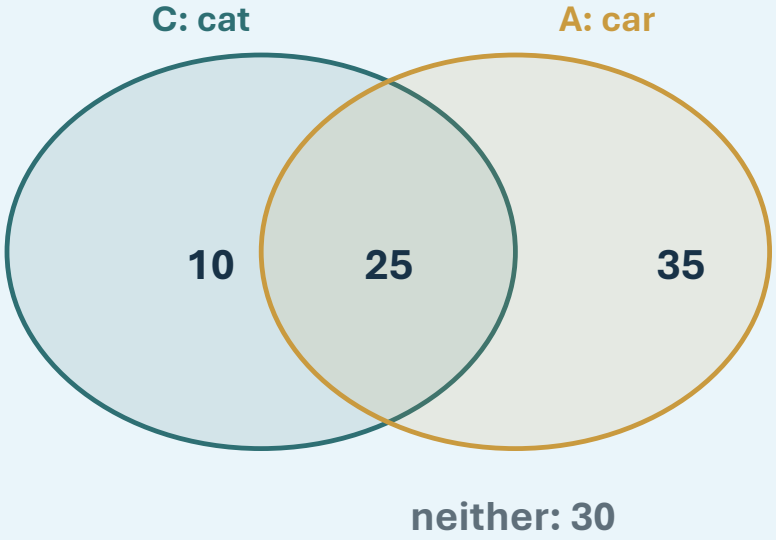
P(C or A)
= P(C) + P(A) − P(C and A)

35/100 + 60/100 − 25/100
= 70/100

The 25 households in the overlap appear in both circle totals.

What would go wrong if we used 35 + 60 = 95?

Complements



P(not A) = 1 − P(A)

P(cat or car) = 70/100

P(neither cat nor car) = 30/100

Complement means everything outside the event.

Disjoint events

TEST

Can both happen for the same randomly selected household?

DISJOINT

exactly 0 cars versus exactly 2 cars

NOT DISJOINT

has a cat versus has a car

If events are disjoint, P(A or B) = P(A) + P(B).

Independent events

Events are independent when knowing one happened does not change the probability of the other.

HOUSEHOLD QUESTION

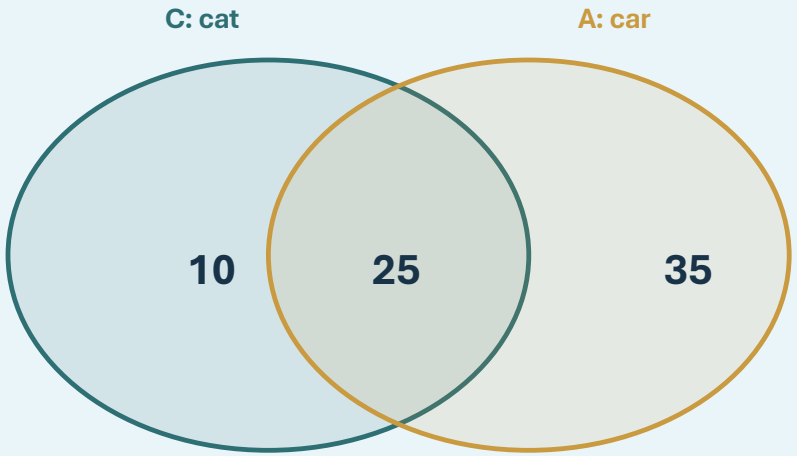
Does knowing a household has a car change how likely it is to have a cat?

KEY DISTINCTION

Disjoint asks whether both can happen. Independent asks whether one changes the probability.

Overlap alone does not answer independence.

Household independence check



P(C) = 0.35
P(A) = 0.60
P(C and A) = 0.25

If independent:

$$0.35 \times 0.60 = 0.21$$

Observed overlap is 0.25, so this constructed example does not fit independence exactly.

Two-way table independence check

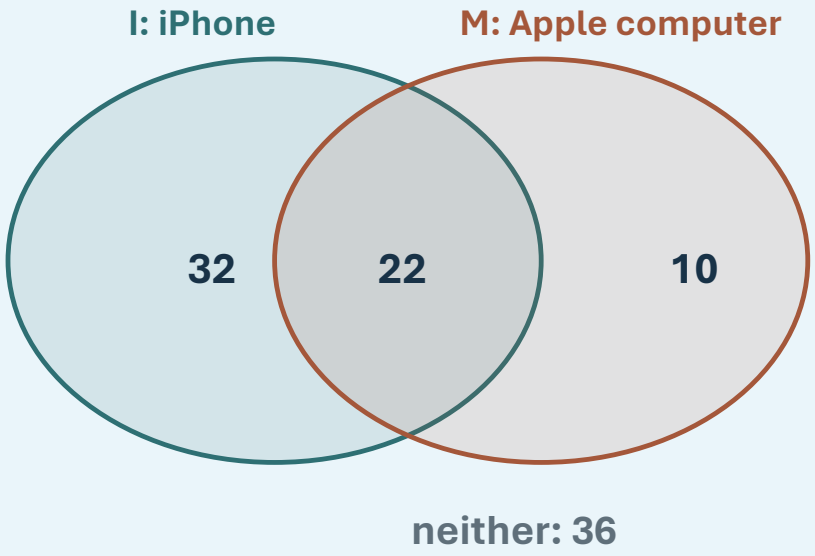
	Car	No car	Total
Cat	25	10	35
No cat	35	30	65
Total	60	40	100

Product check:
P(C and A) = 0.25
P(C)P(A) = 0.21

Conditional check:
P(C | A) = 25/60
P(C) = 35/100

Does knowing the household has a car change the probability of cat ownership?

Phone and computer Venn



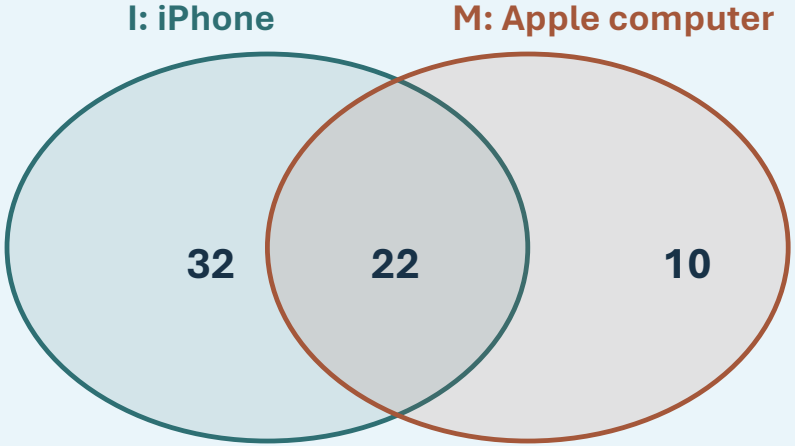
SETUP

100 students
54 use an iPhone
32 use an Apple computer
22 use both

ASK

What is P(iPhone or Apple computer)?
What is P(neither)?

Phone and computer independence



P(I) = 0.54
P(M) = 0.32
P(I and M) = 0.22

Under independence:

$0.54 \times 0.32 = 0.1728$

Two-city rain example

HYPOTHETICAL DAILY WEATHER

P(New York rain) = 0.30
P(Boston rain) = 0.35
P(both rain) = 0.18

$$0.30 \times 0.35 = 0.105$$

0.18 is larger than 0.105, so these hypothetical probabilities do not support independence.

Conditional probability

P(A given B) means: restrict attention to B, then ask how often A occurs.

	Poor sleep	Not poor sleep
High social use	28	42
Lower social use	36	94

Among high-use students, what proportion report poor sleep?

Probability trees

FIRST BRANCH

Choose the first event or group.

SECOND BRANCH

Condition on the first branch, then split into the next outcomes.

PATH PROBABILITY

Multiply along a path. Check that branches add to 1 at each split.

Rule choice map

not A or none

complement

A and B

overlap

A or B

check for disjoint and overlap

given A or among A

conditional denominator

independent

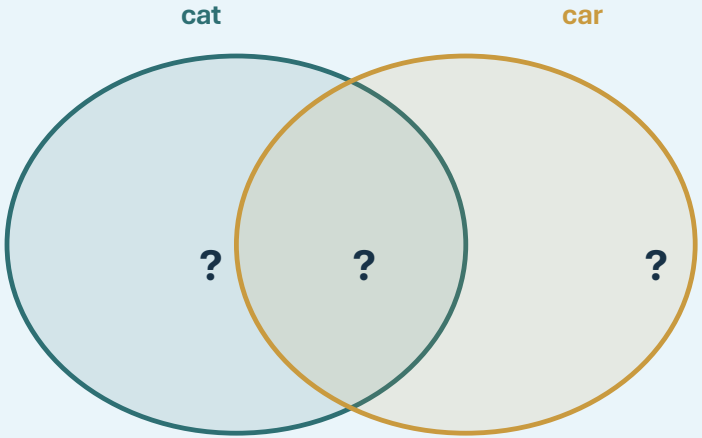
multiplication rule

Read the event wording first. Rewrite the wording into events. The formula comes after the meaning.

Fast practice: households

IN 100 HOUSEHOLDS

45 have a car
30 have a cat
15 have both



DO THIS FIRST

Draw the Venn regions.
Find P(car or cat).
Find P(neither).
Check independence.

Answers: cat only 15, car only 30, both 15, neither 40. Not exactly independent.

Quiz 1 Reminders

Quiz 1: Lectures 1–5 and OpenIntro Chapters 1–2

Today's probability material is not on Quiz 1.

Quiz 1 is paper-based, 15 minutes, at the end of lecture. You may bring one single-sided note.